



Comparison of iodine-125 plaque brachytherapy and gamma knife stereotactic radiosurgery treatment outcomes for uveal melanoma patients

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Abstract

Purpose To compare the efficacies of iodine-125 brachytherapy (IBT) and gamma knife stereotactic radiosurgery (GKRS) in the treatment of posterior uveal melanoma.

Methods The demographic data and tumor characteristics at diagnosis of 201 patients treated with IBT and 52 patients treated with GKRS were recorded. The two treatments were then compared in terms of complications, local control, eye retention, metastasis, and overall survival rate.

Results The median follow-up time was 56 months for the GKRS group and 45 months for the IBT group ($p=0.167$). There were no significant differences in demographic data or tumor characteristics between the groups at diagnosis. Radiation retinopathy, radiation optic neuropathy, and neovascular glaucoma occurred at similar rates in both groups. However, radiation maculopathy and cataracts occurred more frequently in the GKRS group. The number of cases that have developed vision loss (worsening of best-corrected visual acuity on three or more lines on the Snellen chart) was significantly higher in the GKRS group (60%) compared to the IBT group (44%) ($p=0.048$). Local control, metastasis, and 5-year overall survival rates were statistically similar in both groups.

Conclusions GKRS can be preferred as an eye-sparing treatment option for posterior uveal melanoma in cases where brachytherapy cannot be used.

Key messages

- The most common therapy for uveal melanoma is plaque brachytherapy. It has a high success rate as an eye sparing treatment model.
- Stereotactic radiotherapy, which has been recently used for uveal melanoma, provides good results particularly for saving eyes.
- Stereotactic radiotherapy study reports for uveal melanoma from different clinics, reveals different results and complications which need to be more investigated
- This tertiary monocentric study retrospectively compares uveal melanoma treatment outcomes of stereotactic radiotherapy versus iodine plaque brachytherapy in balanced groups regarding tumor size and location.

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Keywords Uveal melanoma · Brachytherapy · Gamma knife · Stereotactic

Abbreviations

UM	Uveal melanoma
IBT	Iodine-125 brachytherapy
GKRS	Gamma knife stereotactic radiosurgery
BCVA	Best-corrected visual acuity
IOP	Intraocular pressure
USG	Ultrasonography
CT	Computed tomography
PET	Positron emission tomography
AJCC	The American Joint Committee on Cancer
RR	Radiation retinopathy
RON	Radiation optic neuropathy
NVG	Neovascular glaucoma
RM	Radiation maculopathy
COMS	The Collaborative Ocular Melanoma Study
SRT	Stereotactic radiotherapy
LINAC	Linear accelerator

Introduction

Uveal melanoma (UM) is the most common primary intraocular malignancy in adults. The mean age-adjusted incidence of UM in the USA is 5.2 per million [1]. Traditionally, UM was treated with enucleation; however, conservative treatments have been preferred in the last 30 years as there is no difference in survival rates between the two approaches [2–5]. Currently, plaque brachytherapy is one of the most commonly used UM treatments [5–10].

Gamma knife stereotactic radiosurgery (GKRS) is another preferred eye-sparing treatment for UM patients. In several retrospective studies, the GKRS treatment results were compared to previously reported results from brachytherapy and GKRS treatments in other clinics [11–15]. However, singular studies comparing these eye-sparing treatment options within a single institution are in insufficient numbers within the literature. In these studies, GKRS treatment results were compared with brachytherapy and GKRS studies in other clinics reported previously.

In this study, we compared the results of IBT and GKRS treatments in UM patients.

Materials and methods

Subjects

We retrospectively analyzed 201 UM patients treated with IBT and 52 UM patients treated with GKRS from January

2000 to December 2019 at the Department of Ophthalmology, Cerrahpasa Medicine Faculty, Istanbul University – Cerrahpasa. The consent of the institutional ethics committee was obtained prior to the study. The study was performed according to the Declaration of Helsinki. Medium and large posterior UM patients were included in the study, and patients with metastasis at diagnosis were exempted. Before treatment, Snellen charts, slit-lamp biomicroscopy, indirect ophthalmoscopy, color fundus photography, and A- and B-scan ultrasonography were used in the ophthalmological examinations of the patients. The demographic data, best-corrected visual acuity (BCVA), intraocular pressure (IOP), tumor thickness, largest tumor basal diameter, and tumor location of each patient were recorded. Post-treatment follow-ups were performed at 1, 3, 6, and 12 months, and every 3 months thereafter. Complete ophthalmological examination, color fundus photography, and A- and B-scan ultrasonography (USG) were performed at each visit. Metastasis was monitored with liver function tests, liver ultrasonography, chest radiography, as-needed computed tomography (CT), and positron emission tomography (PET) every 6 months.

Techniques

In the plaque brachytherapy treatments, an iodine-125 isotope was used. Plaque size was calculated within a 2-mm surgical margin of the tumor's size. A notched plaque was used in juxtapapillary tumors. At the tumor apex, the planned effective dose was approximately 80–100 Gy. The plate was sutured to the episclera at the tumor location while the patient was under general anesthesia. Tumor location was checked by transillumination and indirect ophthalmoscopy. Intraoperative USG was used in some cases. The patients were hospitalized in our clinic during the postoperative period. After the planned effective dose was achieved (approximately 2–7 days), the plaque was removed under general anesthesia.

The Gamma Unit B model of the Leksell stereotactic system (Elekta Instruments, Stockholm, Sweden) was used. Retrobulbar anesthesia was applied to the patients. A Leksell stereotactic head frame was fixed to the outer tabular of the skull with local anesthesia. High-resolution magnetic resonance imaging (MRI) was used as an aid in determining tumor location, treatment planning, and dose planning and for inspect eye shifting during the treatment. The full dose was delivered to the tumor at the 100% isodose line. The prescribed radiation dose at the tumor periphery was standardized to 30 Gy at the 50% isodose field [13].

Statistical analysis

Statistical analysis was performed using SPSS statistics software v25. Descriptive statistics are expressed as mean $\pm$ standard deviation, median, and 25–75. Percentile values depended on the data. Normality of the samples was evaluated by the Kolmogorov–Smirnov test, the Shapiro–Wilk test and Q-Q plot. A chi-square test was used to compare categorical variables, and a Mann–Whitney U test was used to compare continuous variables between groups. Survival assessment was measured with Kaplan–Meier analysis. A log-rank test and Cox regression analysis were performed to evaluate the factor effects on survival. A value of $p < 0.05$ was considered statistically significant.

Results

Demographic data and tumor characteristics are shown in Table 1. There were no significant differences between the two groups in terms of age, gender, laterality, BCVA, tumor size, tumor location, median follow-up time, and the American Joint Committee on Cancer (AJCC) TNM classification before treatment.

After treatment, local recurrence developed in 7 patients (13.4%) in the GKRS group and in 14 patients

(7%) in the IBT group; this difference was not significant ($p = 0.13$). In medium-sized UM, 5 patients (14%) in the GKRS group and 11 patients (6.2%) in the IBT group developed local recurrence. In large-sized UM, 2 patients (12%) in the GKRS group and 3 patients (10%) in the IBT group developed local recurrences. There were no significant differences between the two groups in terms of local recurrence in medium and large UM ($p = 0.103$, $p = 0.82$, respectively). According to the AJCC classification, local recurrence occurred as follows in the GKRS group and in the IBT group, respectively: there were 1 (11%) and 1 (3%) patient in the T1 stage; 2 (10%) and 7 (8%) patients in the T2 stage; and 2 (11%) and 6 (8%) patients in the T3 stage. There were no significant differences between the groups in terms of local recurrence in the T1, T2, and T3 stages ($p = 0.312$, $p = 0.721$, $p = 0.672$, respectively). In the GKRS group, local recurrences developed in the T4 stage in two out of five patients (40%). The groups were not statistically compared since there was only one patient in the T4 stage in the IBT group.

Local control was achieved in 87% of the patients in the GKRS group and in 93% of the patients in the IBT group. There were no significant differences between the two groups; however, eye retention rate was lower in the GKRS group (81%) than in the IBT group (95%) (Table 2). The metastasis rate was 21% in the GKRS group and

Table 1 Patient demographic data and tumor characteristics

	GKRS	IBT	<i>p</i> value
Number of patients, <i>n</i>	52	201	
Age, mean $\pm$ standard deviation	57.6 $\pm$ 12.6	57.5 $\pm$ 13.3	0.976
Gender, male/female	26/26	95/106	0.724
Laterality, right/left	27/25	105/96	0.981
First BCVA (LogMAR), median (range), mm	1.00 (0.22–1.51)	1.00 (0.1–1.51)	0.596
Largest basal diameter, median (range), mm	11.25 (8.87–13.43)	12 (10–14)	0.347
Tumor thickness, median (range), mm	8 (5.75–9.85)	6.5 (5–10)	0.056
AJCC T stage			
T1, <i>n</i> (%)	9 (17%)	33 (17%)	0.877
T2, <i>n</i> (%)	20 (38%)	92 (45%)	0.344
T3, <i>n</i> (%)	18 (35%)	75 (38%)	0.719
T4, <i>n</i> (%)	5 (10%)	1	<0.001
Tumor location			0.942
Choroidal, <i>n</i> (%)	47 (90%)	181 (90%)	
Ciliochoroidal, <i>n</i> (%)	5 (10%)	20 (10%)	
Quadrant locations			
Macula, <i>n</i> (%)	16 (30%)	67 (33%)	0.725
Peripapillary, <i>n</i> (%)	7 (15%)	13 (6%)	0.95
Superior, <i>n</i> (%)	9 (17%)	41 (20%)	0.617
Inferior, <i>n</i> (%)	8 (15%)	35 (17%)	0.728
Nasal, <i>n</i> (%)	6 (11.5%)	24 (12%)	0.936
Temporal, <i>n</i> (%)	6 (11.5%)	24 (12%)	0.936
Follow-up time, months, median (range)	56 (30–90)	45 (30–73)	0.167

15% in the IBT group; this difference was not significant ($p=0.254$). The 5-year overall survival rate was similar in the GKRS (88%) and IBT (89%) groups according to our Kaplan–Meier analysis (log-rank test $p=0.805$) (Fig. 1).

Visual acuity (VA) and complications

In the GKRS group, BCVA at diagnosis was found to be 20/50 or better on the Snellen chart in 18 patients (35%), 20/200–20/50 in 12 patients (23%), and 20/200 or worse in 22 patients (42%). After treatment, the final BCVA was found to be 20/50 or better in 9 patients (17%), 20/200–20/50 in 7 patients (14%), and 20/200 or worse in 36 patients (69%). In the IBT group, BCVA at diagnosis

was 20/50 or better in 51 patients (25%), 20/200–20/50 in 52 patients (26%), and 20/200 or worse in 98 patients (49%). The final BCVA after treatment was 20/50 or better in 42 patients (21%), 20/200–20/50 in 39 patients (19%), and 20/200 or worse in 120 patients (60%). The rate of cases that developed vision loss (worsening of BCVA in three or more lines on the Snellen chart) was significantly higher in the GKRS group (60%) than in the IBT group (4%) ($p=0.048$).

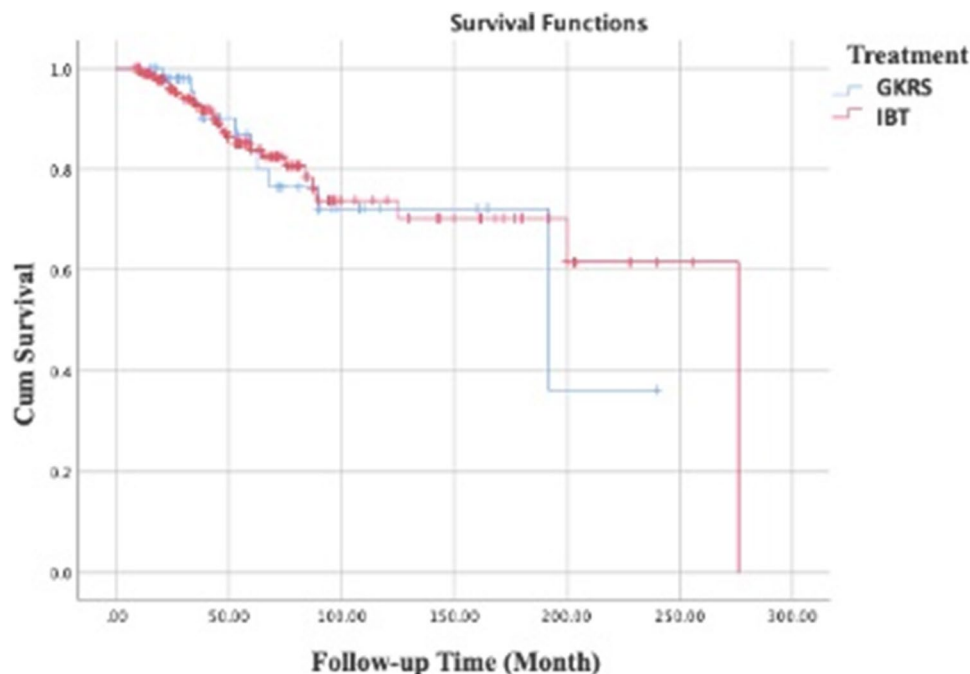
Complications are listed in Table 2. Radiation retinopathy (RR), radiation optic neuropathy (RON), and neovascular glaucoma (NVG) were seen at similar rates in both groups. However, radiation maculopathy (RM) and cataracts occurred more frequently in the GKRS group.

Table 2 Treatment outcomes and complications

	GKRS	IBT	<i>p</i> value
Final BCVA LogMAR, median (range)	1.80 (1.00–3.10)	1.51 (0.7–3.10)	0.46
Vision loss*	31 (60%)	89 (44%)	0.048
Radiation retinopathy (RR), <i>n</i> (%)	25 (48%)	79 (39.3%)	0.251
Radiation maculopathy (RM), <i>n</i> (%)	20 (38%)	47 (23.4%)	0.028
Cataract, <i>n</i> (%)	28 (54%)	70 (35%)	0.012
Radiation optic neuropathy (RON), <i>n</i> (%)	7 (13.4%)	20 (10%)	0.464
Neovascular glaucoma (NVG), <i>n</i> (%)	4 (8%)	14 (7%)	0.855
Local recurrence, <i>n</i> (%)	7 (13%)	14 (7%)	0.13
Eye retention, <i>n</i> (%)	42 (81%)	191 (95%)	<0.001
Metastasis, <i>n</i> (%)	11 (21%)	31 (15%)	0.254
Five-year overall survival rates	88%	89%	0.805

*Worsening of BCVA three or more lines (on the Snellen chart)

Fig. 1 Kaplan–Meier cumulative overall survival analysis of UM patients treated with iodine-125 plaque brachytherapy (IBT) and gamma knife stereotactic radiosurgery (GKRS)



Discussion

Radiotherapy is the most widely used treatment in UM, today. Since the Collaborative Ocular Melanoma Study (COMS) trials, brachytherapy has been widely preferred [4–10]. However, proton beam radiotherapy can also be applied [16–19], and stereotactic radiosurgery can be used when brachytherapy and proton beam radiotherapy are not available options. Zambrano et al. reported the GKRS treatment in UM for the first time [20]. In several published GKRS studies, a local control rate of over 90% has been obtained [11–15, 21]. In previous studies, stereotactic radiotherapy was compared to brachytherapy and proton beam radiotherapy results from other clinics. Fakiris et al. found a local control rate of 94.4% and, comparing these results to those of the COMS study, reported that both studies had similar local control and metastasis rates [12]. In previous studies in our clinic, Sarici and Pazarli found a local control rate of 90% and a 5-year overall survival rate of 86%; this was in the range of previous findings in other radiotherapy trials [13]. However, the groups compared in the above-mentioned studies are heterogeneous, as the studies were conducted in different clinics. Factors such as tumor size, age, gender, ethnic group, and tumor location could affect the prognosis of UM [22–26]. Furthermore, tumor size, radiation dose, and tumor location could affect treatment complications and visual prognosis [27–33]. Tumor size is one of the most important prognostic factors. Increasing tumor thickness is associated with increasing risk for metastasis [25, 26]. Increasing tumor size results in the use of higher radiation dose and increased complication rates. Large tumor size increases radiation retinopathy and secondary glaucoma and led to poor visual prognosis [30–32]. Tumor location is another important prognostic factor. The risk for RM is associated with distance to foveola. Shorter tumor distance to the foveola increases the development of RM [27]. Visual prognosis is worse in macular and juxtapapillary melanoma. For this reason, homogeneous groups formed by eliminating differences in terms of all these prognostic factors will provide a healthier comparison. It will be easier to obtain balanced study groups with a single-center analysis. Comparing two homogeneous groups in terms of demographic data and tumor characteristics will give more reliable results than previous comparisons that used data from different clinics. However, there are few clinics where two different radiotherapy options are applied. For this reason, in UM patients, the number of studies comparing two radiotherapy options with homogenous groups within a single institute is insufficient. There are a few comparative studies of brachytherapy and stereotactic radiotherapy (SRT) [32, 34, 35]. Krema et al. compared

the complication rates of linear accelerator (LINAC) and IBT in juxtapapillary choroidal melanomas; they found NVG (47%, 8%, respectively), RR (89%, 59%, respectively), and RON (74%, 39%, respectively) higher in the SRT (LINAC) group than in the IBT group. They found similar local control rates in both groups [34]. However, a higher dose radiation (70 Gy) was used in SRT compared to our study (30 Gy), and only juxtapapillary choroidal melanomas were included. These procedural variances may be the cause of the differences in complication rates seen in the two studies. Dinca et al. compared the results of using different peripheral doses in UM patients treated with GKRS. They reported that the usage of low doses (35 Gy) did not reduce the survival rates and provided decreased complication rates and good visual prognosis [36]. Siedlecki et al. compared ruthenium-106 brachytherapy and cyber knife SRT treatments in terms of secondary glaucoma. They found no differences in RR between the groups, but the rates of secondary glaucoma were significantly higher and tumor sizes were significantly larger in the SRT group. However, tumor size was higher in the SRT group [32]. In a study conducted by Mazzini et al., secondary glaucoma and radiation retinopathy were found to be higher in the gamma knife group than in the ruthenium-106 brachytherapy group. Although the overall survival rates were found to be similar in both groups, the local control rate was lower (77%) in the gamma knife group. However, tumor sizes were higher in the GKRS group. Additionally, they conducted the study with a small patient population ($n = 48$) [35]. These conditions may have affected the results.

To our knowledge, the present study is the first single-institution comparison of the iodine-125 isotope brachytherapy (IBT) to GKRS for posterior uveal melanoma. In our study, complication, local control, metastasis, and survival rates in the IBT and GKRS groups were within the range of previous findings [5, 8, 10, 12, 13, 27, 36]. We found no significant differences between GKRS and IBT in terms of local control, distant metastasis, and 5-year overall survival rates. NVG was similar in both groups ($p = 0.855$). The reason NVG was found to be higher in SRT groups in previous studies may be due to the large tumor sizes in SRT groups [32, 35]. Tumor size is one of the most important risk factors for the development of secondary glaucoma [30, 32, 37, 38]. There were no differences between the two groups in terms of RR and RON in our study; however, cataracts, and RM were higher in the GKRS group. Additionally, visual prognosis was worse in the GKRS group, which may be due to the group's higher cataract and RM rate.

In conclusion, despite poorer visual prognosis and higher complication rates in GKRS, there were no differences in

terms of survival and local control between GKRS and IBT. GKRS can be used as an alternative to enucleation in the treatment of selected UM patients in cases where brachytherapy cannot be applied. The most important limitation of our study is that it is retrospective. Randomized controlled prospective studies will provide us with additional data.

Data availability All data related to the study are included in the article.

Code availability Not applicable.

Declarations

Ethics approval The consent of the Ethics Committee of Cerrahpasa Medical Faculty, Istanbul University – Cerrahpasa, Turkey, was obtained prior to the study. All procedures performed in studies involving human participants were in accordance with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication All authors warrant that their contribution is original and accept responsibility for releasing this material.

Conflict of interest The authors declare no competing interests.

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